

A Deep Learning Approach to Pan-Arabic Dialect Detection on the IADD Corpus

Ayman Kacan, Mohamad El Dayeh, Tarek Al Saleh

Department of Computer Science and Mathematics

Lebanese American University

ayman.kacan@lau.edu, mohamad.eldayeh01@lau.edu, tarek.alsaleh@lau.edu

Abstract

Arabic Dialect Detection (ADD) is a critical precursor to effective Natural Language Processing (NLP) in the Arab world, yet it remains challenging due to the lack of standardization and the morphological complexity of Dialectal Arabic (DA). This proposal outlines an analytical deep learning approach to automatically classify Arabic text into its geographic dialect. By leveraging the Integrated Arabic Dialect Dataset (IADD), which harmonizes over 135,000 texts across five major regions, we aim to develop a robust classification model capable of generalizing across varied and noisy web-based texts, based on a replication of recent literature in this regard (Saleh et al., 2025).

1 Introduction and Problem Statement

Arabic is a highly diglossic language, characterized by a formal written standard, Modern Standard Arabic (MSA), and numerous spoken dialects that vary significantly by geographic region (Abu Kwaik et al., 2018a). In the context of digital communication, web users frequently write in Dialectal Arabic (DA), which diverges from MSA in syntax, vocabulary, and morphology.

Accurate Dialect Detection is a crucial prerequisite for downstream NLP applications, such as localized sentiment analysis and tailored machine translation. The primary challenge, and the niche we aim to address, is the cross-domain and cross-regional generalization of ADD models. Many existing systems are trained on monolithic datasets representing a single text domain (e.g., exclusively Twitter data) or a narrow geographic region. Consequently, these models exhibit poor robustness when faced with diverse, short-form, and noisy text originating from multiple platforms and regions. Our objective is to achieve a deep learning architecture design that accurately maps varied Arabic web texts to their region of origin.

2 Literature Review

Early approaches to ADD primarily relied on traditional machine learning techniques, such as Support Vector Machines (SVMs) and Naive Bayes, utilizing character and word n-grams as primary features. While establishing strong baselines, these models often fail to capture the complex, long-range semantic dependencies present in dialectal variations.

With the advent of deep learning, researchers transitioned to sequence models like Bidirectional LSTMs (BiLSTMs) and Convolutional Neural Networks (CNNs). Furthermore, the creation of regional corpora propelled localized research. For example, Abu Kwaik et al. (2018b) introduced the SHAMI dataset, comprising over 117,000 sentences focused specifically on the Levantine dialects (Palestine, Jordan, Lebanon, and Syria). While deep learning models trained on SHAMI demonstrated high accuracy for Levantine identification, their scope is inherently restricted.

Recent trends lean heavily on Large Language Models (LLMs) and transformer-based architectures (e.g., AraBERT, MARBERT), though these models require more computational resources. There remains a need for research into lightweight, algorithmic architectures that achieve high accuracy and trustworthiness on Pan-Arabic data without the prohibitive computational overhead of mainstream LLMs. For the sake of this paper, we shall replicate the process of Saleh et al. (2025).

3 Dataset Selection

To train our models, we will utilize the Integrated Arabic Dialect Dataset (IADD) compiled by Zahir (2022). IADD is a large-scale, unified corpus constructed by harmonizing subsets from five distinct open-source corpora: DART, SHAMI, TSAC, PADIC, and AOC.

The dataset contains 135,804 annotated texts

stored in a structured JSON format. It provides a highly diverse training environment by covering five major Arabic dialectal regions: Maghrebi (25%), Levantine ($\approx 64\%$), Gulf ($\approx 5\%$), Egyptian (3.6%) and Iraqi ($< 1\%$). Crucially, for our problem statement, IADD encompasses various forms of noisy digital text, including tweets, Facebook posts, and online newspaper comments. Because SHAMI is a constituent subset of IADD, utilizing this integrated dataset allows us to build upon prior Levantine-focused research while expanding the model’s capability to a comprehensive, Pan-Arabic scale.

References

- Kathrein Abu Kwaik, Motaz Saad, Stergios Chatzikyriakidis, and Simon Dobnik. 2018a. [A Lexical Distance Study of Arabic Dialects](#). *Procedia Computer Science*, 142:2–13. The 4th International Conference on Arabic Computational Linguistics (ACLing 2018).
- Kathrein Abu Kwaik, Motaz Saad, Stergios Chatzikyriakidis, and Simon Dobnik. 2018b. [Shami: A Corpus of Levantine Arabic Dialects](#). In *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)*, Miyazaki, Japan. European Language Resources Association (ELRA).
- Hager Saleh, Alanoud AlMohimeed, Reham Hassan, Maryam M. Ibrahim, Shammir H. Alsamhi, Mohamed R. Hassan, and Sameer Mostafa. 2025. [Advancing Arabic dialect detection with hybrid stacked transformer models](#). *Frontiers in Human Neuroscience*, 19:1498297.
- Jihad Zahir. 2022. [Iadd: An integrated arabic dialect identification dataset](#). *Data in Brief*, 40:107777.